SOA Mini-Project Documentation

Restaurant Management

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Project Overview

The Restaurant App is a full restaurant management system built entirely in Node.js using a microservices architecture. Instead of one big application, the system is split into 6 independent microservices plus an API Gateway, each responsible for one specific domain of the business.
A customer can register, log in, reserve a table or book the whole restaurant for a private event, browse the menu, place a food order, and leave feedback. The restaurant can manage special events like parties , musicals ,and bands…

Services and their ports:

- API Gateway (port 3000) → REST + GraphQL entry point for clients
- User Service (port 50051) → Registration, login, JWT authentication
- Reservation Service (port 50052) → Table and event reservations
- Order Service (port 50053) → Menu and food orders
- Notification Service (port 50054) → sends alerts
- Feedback Service (port 50055) → Customer ratings and comments
- Event Service (port 50056) → Restaurant special events

Asynchronous events via Kafka:

In this project, Kafka is used to send events between services asynchronously, so services can work independently without blocking each other.
Kafka topics in this project:

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Action	published by	consumed by	triggered when
user-registered	User Service	Notification Service	new user registers
reservation-confirmed	Reservation	Notification Service	a reservation is

	Service		created
order-placed	Order service	Notification Service	an order is placed
feedback-submitted	Feedback Service	Notification Service	feedback is submitted
event-created	Event Service	Notification Service	restaurant event is created

technologies used

- Node.js
- gRPC
- Protocol Buffers
- REST
- GraphQL
- Kafka
- SQLite
- RxDB
- Docker and Docker Compose
- bcryptjs
- jsonwebtoken (JWT)

Architecture Diagram